Recoil and explosions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “recoil” and “external” to explain your answer. Sentence starters are optional.

recoil: motion backward when another part is sent forward

external: acting from outside the chosen system

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A cart has momentum **-15 kg·m/s** and mass **3 kg**. **Find velocity** and its new momentum if velocity reverses at unchanged speed.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part’s momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A **1 kg** cart leaves at **+4 m/s**; a **2 kg** cart must have momentum **-4 kg·m/s**, hence velocity **-2 m/s**.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two carts push apart from rest on an ideal track.
1. **Draw their momentum arrows**.
2. How must the two momenta compare?

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



A moves left B moves right

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

An astronaut of mass **60 kg** and a **3 kg** tool are initially at rest relative to a spacecraft. The astronaut pushes the tool so it moves at **+8 m/s** in that frame. **Label the diagram** and **find the astronaut's velocity**. Neglect external impulse.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A projectile's velocity is (**12**, **-16**) m/s. **Find its speed**. What information would locate it?
Label separate horizontal and vertical motion.

Word help: projectile = an object moving after launch under the stated forces

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

07 Exit check

Two fragments of masses **2** and **3 kg** separate from rest. The **2 kg** fragment moves at **+9 m/s**. **Find the other velocity** and total kinetic energy. Organize the known quantities on the before-and-after diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **60 kg** person and **10 kg** board start at rest. After a push, the board moves at **+3 m/s** relative to the ground. **Find the person's ground velocity** and the board's speed relative to the person. Organize the known quantities on the before-and-after diagram.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3